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Improved partitioning of commercial and non-commercial Deep7 bottomfish catch in the MHI

*Hongguang Ma, Toby Matthews, Marc Nadon, John Syslo, Meg Oshima,
Felipe Carvalho*

NOAA Fisheries Pacific Islands Fisheries Science Center

Outline

1. Non-commercial catch for Deep7 bottomfish and Hawaii Marine Recreational Fishing Survey (HMRFS)
2. Catch rate estimation for pure recreational fishers
3. Fishing effort adjustment
4. Summary and conclusion



Deep7 bottomfish in the Main Hawaiian Islands

Biomass Dynamics Model for Deep7 stock assessment

$$B_{T+1} = B_T + R \cdot B_T \left(1 - \left(\frac{B_T}{K} \right)^M \right) - C_T$$


= commercial catch + non-commercial catch

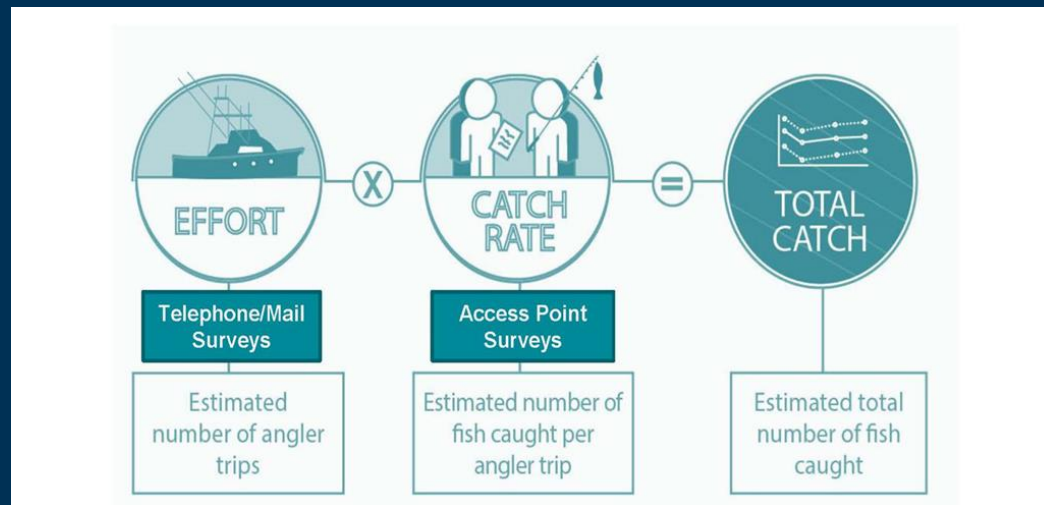


Non-commercial catch estimates for Deep7 bottomfish

Access Point Angler Intercept Survey (APAIS, for catch rate since 2003)

Fishing Effort Survey (FES, for fishing effort since 2018)

Coastal Household Telephone Survey (CHTS, for fishing effort in 2003-2017)



Access Point Angler Intercept Survey

Angler information: type of fishermen

Trip info: fishing area & methods

Catch: angler reported catch & observed catch)

2. ASSIGNMENT NO. ☐ Please indicate if this is your first or second assignment today, by writing "1" or "2"

3. INTERVIEWER ID ☐ ☐ ☐ ☐

4. YR/MO/DAY ☐ ☐ ☐ ☐ ☐ ☐

5. INTERCEPT NO. ☐ ☐

6. INTERVIEW TIME (use 2400 clock) ☐ ☐ ☐ ☐ (Time this interview was completed)

7. STATE CODE ☐ ☐ 8. COUNTY CODE ☐ ☐ 9. SITE CODE ☐ ☐ ☐

10. INTERVIEW STATUS (check mark one) ☐ 1 ☐ 2 ☐ 5 Questionnaire Complete
Refused Non-Key Item
Refused Key Item (Key Item = *)

READ PRIVACY ACT: this study is being conducted in accordance with the privacy act of 1974. You are not required to answer any question that you consider to be an invasion of your privacy.

*11. Would you say you were fishing from a:

1 ☐ Pier/Dock CH 7 ☐ Charter Boat

2 ☐ Jetty, Breakwater, Breachway PR 8 ☐ Private Boat

3 ☐ Bridge, Causeway

4 ☐ Other Man-made Structure (specify) _____

5 ☐ Natural Shoreline (beach, cliffside, etc.).

16. Were you fishing for any particular kinds of fish today? If Yes, what kinds?

☐ No Particular Species / Target Species Code ☐ ☐ ☐

1st Target ☐ ☐ ☐

2nd Target ☐ ☐ ☐

16a. Were you fishing for finfish today?

1 ☐ Yes 2 ☐ No

16b. Did you catch any finfish today?

1 ☐ Yes 2 ☐ No

*12. Was most of your (specify mode) fishing effort today in the:

1 ☐ Ocean

4 ☐ Bay

5 ☐ Other (specify below) _____

Code Q13 as "8" (Not Applicable)

*13. _____ (always for SH mode)

13a. _____

13b. _____

13c. _____

13d. _____

13e. _____

13f. _____

13g. _____

13h. _____

13i. _____

13j. _____

13k. _____

13l. _____

13m. _____

13n. _____

13o. _____

13p. _____

13q. _____

13r. _____

13s. _____

13t. _____

13u. _____

13v. _____

13w. _____

13x. _____

13y. _____

13z. _____

14. What type of GEAR was PRIMARILY used?

01 ☐ Rod & Reel

02 ☐ Hand Line

03 ☐ Hand Pole

04 ☐ Scoop Net

05 ☐ Throw Net

06 ☐ Gill Net

07 ☐ Cross Net

08 ☐ Hukilau Net

09 ☐ Surround Net

10 ☐ Trawl

11 ☐ Trap

12 ☐ Spear

13 ☐ Hand

14 ☐ Crab Net

15 ☐ Other (specify) _____

98 ☐ Unknown

99 ☐ Refused

14a. What METHOD was primarily used?

1 ☐ Trolling

2 ☐ Casting

3 ☐ Bottom Fishing

4 ☐ Hand Lining

Go to Q15a

14b. If bottom fishing or hand lining were you:

1 ☐ Shallow Water (<20 fathoms)

2 ☐ Deep Water (>20 fathoms)

3 ☐ Tuna Hand Lining (hand line only)

15a. To the nearest half-hour, how many hours have you spent (specify mode) fishing today? That is, how many hours have you actually spent with your gear in the water?

☐ ☐ Code 99.9 If Don't Know or Refused

15b. (Only ask if natural shoreline mode [Q11, SH = 5]) How many additional hours do you expect to fish from shore today? That is, how many more hours will you actually have your gear in the water?

☐ ☐ Code 99.9 If Don't Know or Refused

Ever sell any catch?
Sell catch to cover trip expenses or income?

19. Do you ever sell any of the fish you catch?

1 ☐ Yes 2 ☐ No - Go to Q20

19a. When you sell your fish, do you consider yourself a commercial fisherman, making some income or do you sell only to cover your fishing expenses?

1 ☐ For Income 2 ☐ Trip Costs Only - Go to Q20

19b. Do you consider yourself a full-time commercial fisherman?

1 ☐ Yes 2 ☐ No

*20. What is your state & island or county of residence? If county unknown ask: What city or town do you live in?

☐ ☐ State Code; Name _____

☐ ☐ County-Island Code; Name _____

21. What is the zip code of your residence?

☐ ☐ ☐ ☐ ☐ ☐

Code 99997 If Foreign Country; Code 99998 If Don't Know; Code 99999 If Refused

22. Do you live in a private residence, or in some type of housing such as a dormitory, barracks, nursing home or rooming house?

1 ☐ Private Residence 8 ☐ Don't Know

2 ☐ Institutional Housing 9 ☐ Refused

Go to verification box below

23. Does your home have a land line telephone?

1 ☐ Yes 8 ☐ Don't Know

2 ☐ No 9 ☐ Refused

VERIFICATION BOX

In the event my supervisor wishes to verify that I have been conducting interviews here today, may I have your name & phone number?

First Name _____

Last Name _____

☐ Check box if name & phone number not given.

Phone Number (____) _____ - _____

Access Point Angler Intercept Survey

Angler information: type of fishermen

Trip info: fishing area & methods

Catch: angler reported catch & observed catch)

*24. **UNAVAILABLE CATCH.** Did you land any fish that are not here for me to look at? For example, any that you may have thrown back or used for bait? IF YES, COMPLETE TYPE 2 RECORD FOR THIS INDIVIDUAL FISHERMAN, NOT GROUP CATCH. NOTE: FILLETS ARE UNAVAILABLE CATCH.

DISPOSITION CODES FOR Q24

- 1 Thrown back alive / legal
- 2 Thrown back alive / not legal / legality refused
- 3 Eaten / plan to eat
- 4 Used for bait / plan to use for bait
- 5 Sold / plan to sell

- 6 Thrown back dead / plan to throw away
- 7 Some other purpose, Write in margin: _____
- 8 Don't know / Didn't ask
- 9 Refused
- 0 Exchange, Trade

TYPE 2 RECORDS: (INDIVIDUAL CATCH NOT AVAILABLE IN WHOLE FORM)

SPECIES NAME	SPECIES CODE	# OF FISH	DISP	SPECIES NAME	SPECIES CODE	# OF FISH	DISP
				6. _____			
				7. _____			
				8. _____			
				9. _____			
				10. _____			

Reported catch
Unavailable catch

*25. Did you catch any fish I can look at?

- 1 ☐ Yes
- 2 ☐ No – Code Q26, 27, & 28 as "8's," Not Applicable
- 3 ☐ Yes, BUT fish on another angler's form; record intercept no. where fish are listed

Code Q26, 27 & 28 as "8's," Not Applicable

*26. Did you catch these fish yourself or did someone else catch some of them?

- 1 ☐ All caught by fisherman – Code Q27 & 28 as "8's," Not Applicable
- 2 ☐ Other Contributors 8 ☐ Not Applicable

*27. Can you separate out your individual catch?

- 1 ☐ Yes – Code Q28 as "88," Not Applicable
- 2 ☐ No 8 ☐ Not Applicable

*28. How many fishermen including yourself have their catch here? Please do not include anyone who did not catch fish; count only those who have their catch here.

No. of Contributors 88 ☐ Not Applicable

*29. How many people fished on your boat today?

No. of People 88 ☐ Not Applicable or Shoreline Mode

BOX D. If response to Q29 is 01 or Not Applicable/Shoreline Mode, code as "8," Not Applicable or Shoreline Mode. Otherwise, is this the first fisherman from this boat that I have interviewed?

- 1 ☐ Yes 8 ☐ Not Applicable or Shoreline Mode
- 2 ☐ No

Record intercept no. of 1st angler in the fishing party*

*30. **AVAILABLE CATCH.** COMPLETE TYPE 3 RECORD BY ASKING: May I look at your fish? What do you plan to do with the MAJORITY of the catch?

DISPOSITION CODES FOR Q30

- 3 Eaten / plan to eat
- 4 Used for bait / plan to use for bait

- 5 Sold / plan to sell
- 6 Thrown back dead / plan to throw away

- 7 Some other purpose
- 8 Don't know / Didn't ask

- 9 Refused
- 0 Exchange, Trade

TYPE 3 RECORDS: (INDIVIDUAL CATCH AVAILABLE IN WHOLE IDENTIFIABLE FORM)

SPECIES NAME	SPECIES CODE	# OF FISH	PLAN TO SELL	LENGTH (mm)	WEIGHT (kg)	DISP

Observed catch
Available catch

***24. UNAVAILABLE CATCH.** Did you land any fish that are not here for me to look at? For example, any that you may have thrown back or used for bait? IF YES, COMPLETE TYPE 2 RECORD FOR THIS INDIVIDUAL FISHERMAN, NOT GROUP CATCH. NOTE: FILLETS ARE UNAVAILABLE CATCH.

DISPOSITION CODES FOR Q24

- 1 Thrown back alive / legal
- 2 Thrown back alive / not legal / legality refused
- 3 Eaten / plan to eat
- 4 Used for bait / plan to use for bait
- 5 Sold / plan to sell

- 6 Thrown back dead / plan to throw away
- 7 Some other purpose, Write in margin: _____
- 8 Don't know / Didn't ask
- 9 Refused
- 0 Exchange, Trade

Catch disposition

TYPE 2 RECORDS: (INDIVIDUAL CATCH NOT AVAILABLE IN WHOLE FORM)

SPECIES NAME	SPECIES CODE	# OF FISH	DISP	SPECIES NAME	SPECIES CODE	# OF FISH	DISP
1. _____				6. _____			
2. _____				7. _____			
3. _____				8. _____			

***30. AVAILABLE CATCH.** COMPLETE TYPE 3 RECORD BY ASKING: May I look at your fish? What do you plan to do with the MAJORITY of the catch?

Catch disposition

DISPOSITION CODES FOR Q30

- 3 Eaten / plan to eat
- 4 Used for bait / plan to use for bait
- 5 Sold / plan to sell
- 6 Thrown back dead / plan to throw away
- 7 Some other purpose
- 8 Don't know / Didn't ask
- 9 Refused
- 0 Exchange, Trade

TYPE 3 RECORDS: (INDIVIDUAL CATCH AVAILABLE IN WHOLE IDENTIFIABLE FORM)

SPECIES NAME	SPECIES CODE	# OF FISH	PLAN TO SELL	LENGTH (mm)	WEIGHT (kg)	DISP
Observed catch						
Available catch						

Angler profile information

Pure recreational: don't sell any catch

Recreational expense: sell catch to cover trip expenses

Commercial: sell catch for income (part-time and ~~full time~~)

19. Do you ever sell any of the fish you catch?

1 ☐ Yes

2 ☐ No - *Go to Q20*

19a. When you sell your fish, do you consider yourself a commercial fisherman, making some income or do you sell only to cover your fishing expenses?

1 ☐ For Income

2 ☐ Trip Costs Only - *Go to Q20*

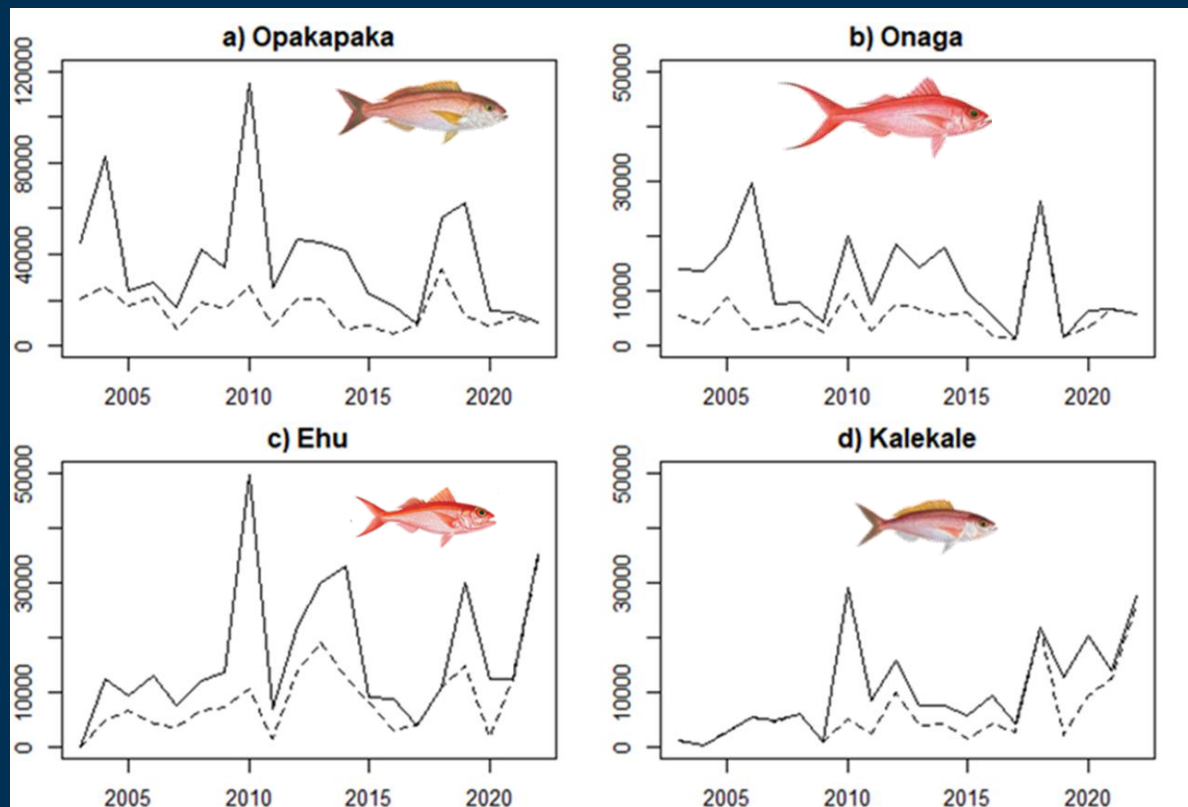
19b. Do you consider yourself a full-time commercial fisherman?

~~1 ☐ Yes~~

2 ☐ No



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*30. **AVAILABLE CATCH (TYPE 3).** INDIVIDUAL OR GROUP CATCH (PR ONLY) AVAILABLE IN WHOLE (ETC). May I look at your fish? What do you plan to do with the MAJORITY of the catch?

DISPOSITION CODES FOR TYPE 3

- | | | |
|------------------------|---------------------------|---------------------------|
| 3 Plan to eat | 5 Plan to sell | 7 Some other purpose |
| 4 Plan to use for bait | 6 Plan to throw away | 8 Don't know / Don't plan |

*24. **UNAVAILABLE CATCH.** Did you land any fish that are not available for sale? COMPLETE TYPE 2 RECORD FOR THIS INDIVIDUAL FISH

- | |
|--|
| 1 Thrown back alive / legal |
| 2 Thrown back alive / not legal / legality refused |
| 3 Eaten / plan to eat |
| 4 Used for bait / plan to use for bait |
| 5 Sold / plan to sell |

Outline

1. Non-commercial catch estimation for Deep7 bottomfish and Hawaii Marine Recreational Fishing Survey (HMRFS)
- 2. Catch rate estimation for pure recreational fishers**
- 3. Fishing effort adjustment**
4. Summary and conclusion



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Angler profile information (for catch rate)

Pure recreational: don't sell any catch (unsold catch)

Recreational expense: sell catch to cover trip expenses (may have CML)

Commercial: sell catch for income (should have CML)

19. Do you ever sell any of the fish you catch?

1 ☐ Yes 2 ☒ No - *Go to Q20*

19a. When you sell your fish, do you consider yourself a commercial fisherman, making some income or do you sell only to cover your fishing expenses?

1 ☐ For Income 2 ☐ Trip Costs Only - *Go to Q20*

19b. Do you consider yourself a full-time commercial fisherman?

1 ☐ Yes 2 ☐ No

Catch rate estimation

$$\text{Catch rate} = \frac{\sum(\text{unsold catch})}{\sum(\# \text{ interviews or contributors})}$$

$\sum(\text{unsold catch})$: catch interviewed from pure recreational fishers

$\sum(\# \text{ interviews or contributors})$: # of interviews or contributors from pure recreational fishers



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Angler profile information from CHTS (for fishing effort)

You never sell any of your catch (**pure recreational**)

You sometimes sell fish to help cover fishing expenses

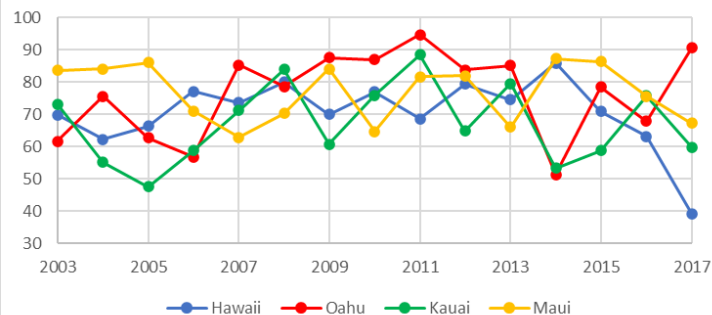
You sell fish for profit to pay your living expenses



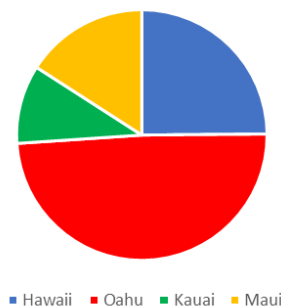
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Fishing effort adjustment for pure recreational fishing

Percent of pure recreational fishing trips by county (Pi)

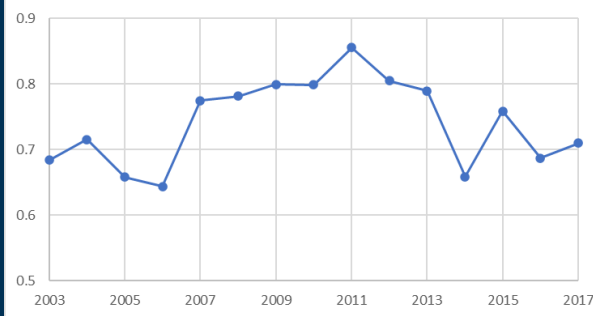


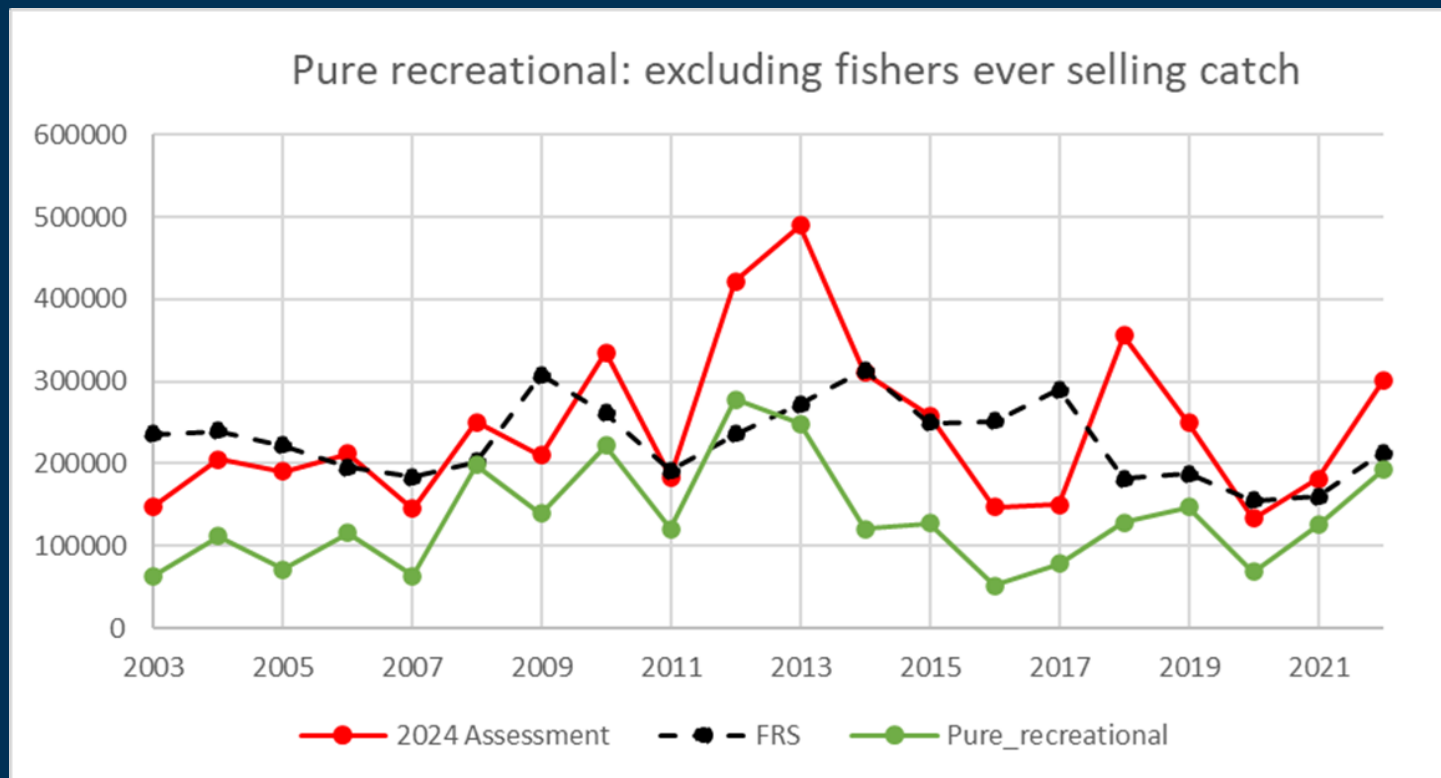
Proportion of fishing trips from each county (Ti)



$$\sum_{n=1}^4 (P_i * T_i)$$

Adjustment





Survey catch from potential CML holders was excluded for catch rate estimation (unsold catch only) and fishing effort was adjusted for pure recreational fishing trips

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Summary and conclusion

The fishing trips from pure recreational fishers accounted for ~75% of the estimated fishing effort from CHTS/FES

The estimated catch from pure recreational fishers was ~55% of the non-commercial catch previously estimated

Separate catch rate and fishing effort estimation by fishing method (bottom fishing vs other methods) or by county may be explored in the future to further improve the catch estimation for Deep7 bottomfish



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